# Research Gap

**Research Gaps in Steroid Detection in Milk Production**

**1. Lack of Real-Time Detection Systems**

Most current techniques (like chromatography and immunoassays) are:

- Time-consuming
- Lab-dependent

**Gap:**
There is a need for **real-time, on-site detection systems** (e.g., biosensors, IoT-based devices) that can be used directly at dairy farms.

**2. Limited Sensitivity for Trace-Level Detection**

Steroids are often present in **very low concentrations (ppb or ppt levels)**.

**Gap:**
Existing methods sometimes fail to detect **ultra-trace levels**, especially in complex milk matrices.

**3. High Cost and Technical Complexity**

Techniques such as:

- HPLC
- LC-MS/MS

require:

- Expensive equipment
- Skilled personnel

**Gap:**
There is a need for **low-cost, user-friendly detection methods** suitable for small-scale dairy farmers.

### 4. Poor Portability of Detection Technologies

Most systems are confined to laboratories.

**Gap:**
Development of **portable devices (handheld sensors)** is still limited and not widely commercialized.

### 5. Limited Use of AI/ML for Detection

Although machine learning is growing, its use in this field is still emerging.

**Gap:**

- Lack of **AI-based predictive models** for steroid contamination
- Minimal integration of **ML with sensor data**

### 6. Inadequate Multi-Residue Detection

Milk may contain multiple contaminants:

- Steroids
- Antibiotics
- Hormones

**Gap:**
Most studies focus on **single-compound detection**, not **multi-residue simultaneous detection**.

### 7. Insufficient Data on Long-Term Exposure Effects

Research often focuses on detection rather than impact.

**Gap:**

- Limited studies on **chronic health effects** of low-level steroid consumption
- Lack of **population-based risk assessment models**

### 8. Weak Regulatory Monitoring Systems

In many regions:

- Monitoring is inconsistent
- Testing frequency is low

**Gap:**
Need for **automated monitoring systems** integrated with regulatory frameworks.

### 9. Milk Matrix Interference Issues

Milk is a complex mixture of:

- Fats
- Proteins
- Enzymes

**Gap:**
These components interfere with detection accuracy, and **robust sample preparation methods** are still lacking.

### 10. Lack of Standardized Detection Protocols

Different studies use different methods.

**Gap:**
No **globally standardized protocols** for steroid detection in milk, leading to inconsistent results.

### 11. Limited Research in Developing Countries

Most studies are concentrated in developed regions.

**Gap:**

- Lack of **region-specific studies** (e.g., India dairy sector)
- Need for **locally adaptable technologies**

**12. Stability and Degradation Studies Are Limited**

Steroids may degrade during:

- Storage
- Processing (boiling, pasteurization)

**Gap:**

Insufficient research on how **processing affects steroid residues**.